

Variationally Consistent Steklov Merging for Modal Poisson Solves on Tetrahedra

(with HDG/Nitsche-type stabilization in moment space + stenciling, mult-by ops and some other stuff)

Sachin Natesh

1 Overview and key principle

This note documents the discrete method implemented in two codes: (i) a robust single-tetrahedron Poisson solver using a stacked overdetermined system solved by sparse QR, and (ii) a two-tetrahedron merge constructed by eliminating volume coefficients and solving a face-only interface problem. The central insight is:

The merge must be the Euler–Lagrange condition of the same discrete functional that defines the stable single-tetrahedron solve.

A naive Steklov merge (imposing only “flux continuity” via an unstabilized DtN map) fails because it does *not* correspond to stationarity of the underlying stacked least-squares functional; it produces a rank-degenerate (or numerically rank-degenerate) interface operator with a large near-nullspace. A variationally consistent stabilization fixes this by penalizing the mismatch between the *induced* trace (produced by the local least-squares extension) and the *skeleton* trace (the global interface unknown).

2 Geometry and bases

Let $\widehat{K}$, K be the reference and physical tetrahedron, respectively. We have the standard affine map

$$x = \Phi_K(\widehat{x}) = V_0 + J_K \widehat{x}, \quad J_K \in \mathbb{R}^{3 \times 3}.$$

Faces are denoted $\widehat{F}$ and $F = \Phi_K(\widehat{F})$. For a face F , let n_F be the outward unit normal.

2.1 Volume basis

Fix polynomial degree n . Let

$$V_n(K) = \mathbb{P}_n(K), \quad M = \dim V_n(K),$$

with modal basis $\{\phi_j\}_{j=1}^M$ (Jacobi-type on the tetrahedron). We represent the discrete solution as

$$u_h(x) = \sum_{j=1}^M c_j \phi_j(x), \quad c \in \mathbb{R}^M.$$

2.2 Face test/trace basis (moment space)

On a face F , define

$$\Lambda_n(F) = \mathbb{P}_n(F), \quad k_f = \dim \Lambda_n(F),$$

with face test basis $\{\psi_i\}_{i=1}^{k_f}$ (Jacobi-type on the triangle).

The *face moment (trace) vector* associated with u_h is

$$\lambda_F(c) \in \mathbb{R}^{k_f}, \quad (\lambda_F(c))_i = \int_F \psi_i u_h ds.$$

This is linear in c , so there exists a matrix $T_F \in \mathbb{R}^{k_f \times M}$ such that

$$\lambda_F(c) = T_F c, \quad (T_F)_{ij} = \int_F \psi_i \phi_j ds.$$

Stacking all four faces of a tetrahedron K ,

$$T_{\text{full}} c = \lambda, \quad T_{\text{full}} \in \mathbb{R}^{(4k_f) \times M}, \quad \lambda = (\lambda_{F_0}, \dots, \lambda_{F_3}).$$

3 Promoted interior PDE constraints and their weights

Let $W_{n-2}(K) = \mathbb{P}_{n-2}(K)$ and choose an interior test basis $\{q_i\}_{i=1}^m$, $m = \dim W_{n-2}(K)$. The method uses *promoted* constraints of the form

$$\int_K (\Delta u_h + f) q_i w_1(x) dx = 0, \quad i = 1, \dots, m. \quad (1)$$

Discrete interior operator. Define $L_{\text{int}} \in \mathbb{R}^{m \times M}$ and $f_{\text{int}} \in \mathbb{R}^m$ by

$$(L_{\text{int}})_{ij} = \int_K (\Delta \phi_j) q_i w_1(x) dx, \quad (f_{\text{int}})_i = \int_K f q_i w_1(x) dx, \quad (2)$$

where $w_1(x)$ is the weight function under which the promoted basis is orthonormal. Then (1) becomes

$$L_{\text{int}} c = -f_{\text{int}}. \quad (3)$$

Remark 1 (Weights and quadrature). *In practice, the integrals in (2) are computed by quadrature on $\hat{K}$ and mapped to K . The promoted operator is assembled in the same moment/test basis as used to project the source f . That is, f_{int} is not arbitrary; it is the vector of moments of f against $\{q_i\}$ with the physical volume measure dx . This basis/weight consistency is crucial for polynomial exactness in the manufactured-solution tests.*

4 The single-tetrahedron solve as a variational problem

The stable single-tet method is defined by minimizing the stacked residual functional

$$\mathcal{J}_K(c; \lambda, f) = \frac{1}{2} \|T_{\text{full}} c - \lambda\|_2^2 + \frac{1}{2} \|L_{\text{int}} c + f_{\text{int}}\|_2^2. \quad (4)$$

Equivalently,

$$\mathcal{J}_K(c; \lambda, f) = \frac{1}{2} \|Ac - b(\lambda, f)\|_2^2, \quad A = \begin{bmatrix} T_{\text{full}} \\ L_{\text{int}} \end{bmatrix}, \quad b = \begin{bmatrix} \lambda \\ -f_{\text{int}} \end{bmatrix}. \quad (5)$$

The computed coefficients are

$$c^*(\lambda, f) = \arg \min_{c \in \mathbb{R}^M} \mathcal{J}_K(c; \lambda, f). \quad (6)$$

4.1 QR construction of the solution map

The code computes c^* using sparse QR (SuiteSparseQR) of the (sparse) stacked matrix A . Let

$$AE = QR,$$

where E is a column permutation, Q has orthonormal columns, and R is upper triangular on the numerical rank. Then the minimum-norm least-squares solution is obtained by

$$c^* = E \begin{bmatrix} R_{11}^{-1}(Q^\top b)_{1:r} \\ 0 \end{bmatrix}, \quad (7)$$

with numerical rank r (matching the code's SPQR bookkeeping).

Linearity. Because c^* is computed by a fixed linear pseudoinverse A^+ ,

$$c^*(\lambda, f) = A^+ \begin{bmatrix} \lambda \\ -f_{\text{int}} \end{bmatrix},$$

hence c^* is affine in the imposed boundary moments λ and in the interior forcing moments f_{int} .

5 Flux moments and trace moments on a chosen face

Fix a face $\Gamma \subset \partial K$. Define

$$T_\Gamma \in \mathbb{R}^{k_f \times M} \quad (\text{trace moment map on } \Gamma), \quad F_\Gamma \in \mathbb{R}^{k_f \times M} \quad (\text{flux moment map on } \Gamma).$$

The *induced trace* (from volume coefficients) is

$$\lambda_\Gamma^{\text{ind}}(c) = T_\Gamma c,$$

and the *flux moments* are

$$\mu_\Gamma(c) = F_\Gamma c, \quad (F_\Gamma)_{ij} = \int_\Gamma \psi_i (\nabla \phi_j \cdot n_\Gamma) ds.$$

(Importantly, F_Γ must be built using the physical gradient transform J_K^{-T} and the physical unit normal; this is why the final corrected implementation builds F_Γ in each leaf, not by scaling a reference precompute.)

6 Local affine response operators (precise construction)

Consider one tetrahedron K and one face Γ . We split the boundary moments into:

$$\lambda = (\lambda_\Gamma, \lambda_{\text{ext}}),$$

where λ_{ext} collects the remaining three face blocks.

Fix $(\lambda_{\text{ext}}, f_{\text{int}})$ and view the solution as a function of λ_Γ only.

6.1 Block injection and solve-many

Let $B_\Gamma \in \mathbb{R}^{(4k_f+m) \times k_f}$ be the injection matrix that writes a k_f -vector into the Γ rows of the stacked right-hand side (and zeros elsewhere). In code this is `B_face(iface)`.

Define the particular solve (interface set to zero):

$$c_0 = c^*(\lambda_\Gamma = 0, \lambda_{\text{ext}}, f_{\text{int}}). \quad (8)$$

Define the homogeneous response matrix

$$C_\Gamma = c^*(\lambda_\Gamma = I, \lambda_{\text{ext}} = 0, f_{\text{int}} = 0) = A^+ B_\Gamma \in \mathbb{R}^{M \times k_f}, \quad (9)$$

computed by applying the same QR solve to the multi-right-hand-side matrix B_Γ .

Then the general solution is

$$c^*(\lambda_\Gamma, \lambda_{\text{ext}}, f_{\text{int}}) = C_\Gamma \lambda_\Gamma + c_0. \quad (10)$$

6.2 Derived face operators

From (10),

$$\mu_\Gamma(\lambda_\Gamma) = F_\Gamma c^* = (F_\Gamma C_\Gamma) \lambda_\Gamma + F_\Gamma c_0 =: S_F \lambda_\Gamma + g_F, \quad (11)$$

$$\lambda_\Gamma^{\text{ind}}(\lambda_\Gamma) = T_\Gamma c^* = (T_\Gamma C_\Gamma) \lambda_\Gamma + T_\Gamma c_0 =: S_T \lambda_\Gamma + g_T. \quad (12)$$

These are the *local affine response operators*:

$$S_F \in \mathbb{R}^{k_f \times k_f}, \quad g_F \in \mathbb{R}^{k_f}, \quad S_T \in \mathbb{R}^{k_f \times k_f}, \quad g_T \in \mathbb{R}^{k_f}.$$

Remark 2 (Interpretation of S_T). *In a classical Dirichlet solve one would have $\lambda_\Gamma^{\text{ind}}(\lambda_\Gamma) = \lambda_\Gamma$, i.e. $S_T = I$ and $g_T = 0$. In the present method, c^* is a least-squares extension balancing interior and trace residuals, so in general $S_T \neq I$. This mismatch is precisely what must be accounted for in a variationally consistent merge.*

7 Two-tetrahedron setting and skeleton vs induced trace

Let A and B be two tetrahedra sharing an interface face Γ . Let $\lambda \in \mathbb{R}^{k_f}$ denote the *skeleton trace moments* on Γ , i.e. the global interface unknown.

For each element, solving the local problem with skeleton input λ produces a volume coefficient vector $c_A(\lambda)$ and $c_B(\lambda)$, hence an *induced trace* on Γ :

$$\lambda_A^{\text{ind}}(\lambda) = T_\Gamma^A c_A(\lambda), \quad \lambda_B^{\text{ind}}(\lambda) = T_\Gamma^B c_B(\lambda).$$

In a perfectly conforming method these would equal λ identically; in the stacked LS method they do not unless λ lies in a compatible subspace.

8 Why the naive Steklov merge is ill-conditioned / rank-degenerate

The naive merge enforces physical flux continuity using the (unstabilized) DtN maps:

$$\mu_A(\lambda) + \mu_B(\lambda) = 0 \iff (S_F^A + S_F^B) \lambda = -(g_F^A + g_F^B). \quad (13)$$

8.1 Mechanism: near-nullspace from smoothing

The local least-squares extension $c^*(\lambda)$ is a discrete harmonic extension operator in high-order face modes: high-degree face moments can be extended into the volume with very small interior energy, producing very small normal flux. Thus S_F has many small singular values and may be numerically rank-deficient:

$$S_F \approx 0 \quad \text{on a large subspace of face modes.}$$

Consequently,

$$K_{\text{naive}} := S_F^A + S_F^B$$

inherits a large near-nullspace and becomes severely ill-conditioned (or numerically rank-deficient). This is exactly what the observed spectra show: a smooth decay followed by a cluster at machine precision.

8.2 Mismatch with the discrete variational structure

More fundamentally, (13) is *not* the stationarity condition of any reduced functional obtained by eliminating c_A, c_B from the true discrete objective (4). As a result, it does not select the interface trace consistent with the local least-squares extension, leading to loss of accuracy (observed plateau).

9 Variationally consistent stabilization (HDG/Nitsche in moment space)

9.1 Augmented numerical flux as a penalty on induced–skeleton mismatch

We define the augmented (numerical) flux moments on Γ for each element:

$$\hat{\mu}(\lambda) = \mu(\lambda) + \tau M_\Gamma (\lambda^{\text{ind}}(\lambda) - \lambda), \quad (14)$$

where M_Γ is the face mass matrix in the same moment basis:

$$(M_\Gamma)_{ij} = \int_\Gamma \psi_i \psi_j \, ds.$$

Here $\tau > 0$ penalizes the mismatch between the induced trace and the skeleton trace in the $L^2(\Gamma)$ moment metric.

Using (11)–(12),

$$\hat{\mu}(\lambda) = (S_F + \tau M_\Gamma (S_T - I))\lambda + (g_F + \tau M_\Gamma g_T). \quad (15)$$

9.2 Stabilized merge equation

Flux continuity is imposed on the augmented flux:

$$\hat{\mu}_A(\lambda) + \hat{\mu}_B(\lambda) = 0. \quad (16)$$

This yields the stabilized interface system

$$K_{\text{stab}}\lambda = -h_{\text{stab}}, \quad (17)$$

with

$$K_{\text{stab}} = (S_F^A + \tau M(S_T^A - I)) + (S_F^B + \tau M(S_T^B - I)), \quad (18)$$

$$h_{\text{stab}} = (g_F^A + \tau M g_T^A) + (g_F^B + \tau M g_T^B). \quad (19)$$

(When element face bases differ, moment vectors are mapped by an orthonormal face change-of-basis P ; the above is written in a common coordinate convention.)

10 Discrete functional for two tetrahedra and Euler–Lagrange derivation

10.1 Global discrete objective (skeleton-coupled)

Define the global functional

$$\begin{aligned} \mathcal{J}(c_A, c_B, \lambda) = & \frac{1}{2} \|T_A c_A - \lambda_A^{\text{data}}\|^2 + \frac{1}{2} \|L_A c_A + f_A\|^2 + \frac{1}{2} \|T_B c_B - \lambda_B^{\text{data}}\|^2 + \frac{1}{2} \|L_B c_B + f_B\|^2 \\ & + \frac{\tau}{2} \|\Pi_\Gamma T_\Gamma^A c_A - \lambda\|_M^2 + \frac{\tau}{2} \|\Pi_\Gamma T_\Gamma^B c_B - \lambda\|_M^2. \end{aligned} \quad (20)$$

Here:

- T_A, T_B are the full face-trace operators for each tet (all faces).
- $\lambda_A^{\text{data}}, \lambda_B^{\text{data}}$ encode the prescribed exterior Dirichlet moments (with zeros on the interface block).
- Π_Γ extracts the interface face block from the stacked face vector (equivalently, we can use T_Γ directly).
- $\|v\|_M^2 := v^\top M v$ is the face-mass norm on Γ .

Meaning. The new terms enforce that the induced trace on the interface is close to the skeleton trace λ . They are the *only* modification required to make hybridization variationally consistent with the local least-squares solves.

10.2 Elimination of volume variables and reduced functional

For a fixed λ , define

$$(c_A(\lambda), c_B(\lambda)) = \arg \min_{c_A, c_B} \mathcal{J}(c_A, c_B, \lambda).$$

The reduced functional is

$$\mathcal{J}_{\text{red}}(\lambda) = \mathcal{J}(c_A(\lambda), c_B(\lambda), \lambda).$$

10.3 Euler–Lagrange equation for λ

The Euler–Lagrange equation for λ is the stationarity condition

$$\nabla_\lambda \mathcal{J}_{\text{red}}(\lambda) = 0.$$

Since λ appears only in the two penalty terms in (20), and those terms are quadratic, we can compute the derivative directly (envelope theorem: derivatives through $c_A(\lambda), c_B(\lambda)$ vanish at the minimizer for variations in λ).

Let

$$r_A(\lambda) = \Pi_\Gamma T_\Gamma^A c_A(\lambda) - \lambda, \quad r_B(\lambda) = \Pi_\Gamma T_\Gamma^B c_B(\lambda) - \lambda.$$

Then

$$\mathcal{J}_{\text{red}}(\lambda) = (\text{terms independent of } \lambda) + \frac{\tau}{2} r_A(\lambda)^\top M r_A(\lambda) + \frac{\tau}{2} r_B(\lambda)^\top M r_B(\lambda).$$

Differentiating with respect to λ gives

$$\nabla_\lambda \mathcal{J}_{\text{red}}(\lambda) = -\tau M r_A(\lambda) - \tau M r_B(\lambda).$$

Setting this to zero yields

$$M(\Pi_\Gamma T_\Gamma^A c_A(\lambda) - \lambda) + M(\Pi_\Gamma T_\Gamma^B c_B(\lambda) - \lambda) = 0. \quad (21)$$

From trace balance to augmented flux balance. The stabilized method implemented in code uses *augmented flux continuity* (16). To see equivalence, note that each local minimizer $c_K(\lambda)$ satisfies the optimality condition for c :

$$(T_{\text{full}}^K)^\top (T_{\text{full}}^K c_K - \lambda_K^{\text{data}}) + (L_{\text{int}}^K)^\top (L_{\text{int}}^K c_K + f_K) + \tau (T_\Gamma^K)^\top M (T_\Gamma^K c_K - \lambda) = 0.$$

This is the discrete analogue of a Nitsche/HDG formulation, and the term

$$\tau (T_\Gamma^K)^\top M (T_\Gamma^K c_K - \lambda)$$

is precisely the penalty contribution that, when interpreted on the face, produces the additional term $\tau M(\lambda^{\text{ind}} - \lambda)$ in the numerical flux (14). In other words, the interface condition (21) is equivalent to continuity of the augmented flux, because the discrete weak form on each element couples normal flux and trace mismatch in the same moment metric.

Operationally, after elimination, the resulting reduced linear system for λ is exactly (17) because

$$\mu(\lambda) = S_F \lambda + g_F, \quad \lambda^{\text{ind}}(\lambda) = S_T \lambda + g_T,$$

and substituting into (14) yields (15), hence (17).

Theorem 1 (Variational-discrete consistency). *The stabilized interface equation (17) is the stationarity condition of the reduced functional $\mathcal{J}_{\text{red}}(\lambda)$ induced by the global discrete functional (20). Equivalently, the merge respects the same discrete variational structure as the local least-squares solves.*

11 Effect of stabilization: coercivity and conditioning

11.1 Coercivity intuition

The penalty term adds control of trace mismatch:

$$\frac{\tau}{2} \|\lambda^{\text{ind}}(\lambda) - \lambda\|_M^2.$$

This removes directions in trace space for which the unstabilized DtN map has nearly zero flux response. Those directions were responsible for the near-null singular spectrum of the naive K .

11.2 Spectral effect (why K becomes well-conditioned)

Write the stabilized operator (one element) as

$$K_K(\tau) = S_F^K + \tau M(S_T^K - I).$$

In the ideal conforming case, $S_T^K = I$ and the penalty vanishes, recovering the classical DtN merge. In the stacked LS method, $S_T^K - I$ is nontrivial and supplies exactly the missing coercivity on the problematic trace modes.

Empirically (and consistent with the observed SVD plots), the naive operator has many small singular values because S_F^K is compact/smoothing on high modes. The penalty term contributes an $O(\tau)$ component in the M -metric on those modes, lifting the spectrum and eliminating numerical rank deficiency.

12 Choice of τ

A robust scaling is

$$\tau = C \frac{(n+1)^2}{h_\Gamma},$$

where h_Γ is a characteristic face diameter (e.g. maximum edge length) and C is a moderate constant.

13 Conclusion

The method is a hybrid Steklov/HDG scheme in moment space: local solves are defined by a stacked least-squares functional, and the interface unknown is obtained by enforcing continuity of an augmented numerical flux that penalizes induced-skeleton trace mismatch. The decisive principle is variational-discrete consistency: the merge must be the Euler–Lagrange equation of the same discrete functional that defines the stable single-tet solve.

Block sparsity assumptions on Jacobi simplex operators

Let

$$\Pi_n^D = \bigoplus_{j=0}^n \mathcal{H}_j^D$$

denote the decomposition of total-degree polynomials into homogeneous subspaces $\mathcal{H}_j^D$ of degree exactly j . We fix, for each j , a Jacobi simplex basis $\{\phi_\alpha^{(\kappa)} : |\alpha| = j\}$ of $\mathcal{H}_j^D$, ordered by a graded rule. We assume the existence of a canonical bijection between multi-indices α with $|\alpha| = j$ and local indices in $\mathcal{H}_j^D$.

All operators below are represented in this basis.

(i) Degree-lowering operators (DMat). Let

$$\mathcal{D} : \Pi_n^D \rightarrow \Pi_{n-1}^D$$

be a linear operator such that

$$\mathcal{D} \mathcal{H}_j^D \subset \mathcal{H}_{j-1}^D, \quad j \geq 1.$$

Then the matrix of $\mathcal{D}$ has block structure

$$\mathcal{D} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ D_1 & 0 & 0 & \cdots & 0 \\ 0 & D_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_n & 0 \end{pmatrix},$$

where $D_j : \mathcal{H}_j^D \rightarrow \mathcal{H}_{j-1}^D$.

Sparsity assumption. There exists a finite set

$$\Delta_{\mathcal{D}} \subset \mathbb{Z}^D, \quad \sum_{d=1}^D \Delta_d = -1,$$

independent of j , such that

$$\mathcal{D}\phi_{\alpha}^{(\kappa)} \in \text{span}\{\phi_{\alpha+\Delta}^{(\kappa')} : \Delta \in \Delta_{\mathcal{D}}\}.$$

(ii) Degree-filtering operators (KMat). Let

$$\mathcal{K} : \Pi_n^D \rightarrow \Pi_n^D$$

be a linear operator arising from a *natural promotion* of Jacobi parameters,

$$\kappa_{\text{tgt}} = \kappa_{\text{src}} + e_r, \quad r \in \{0, \dots, D\},$$

corresponding to multiplication of the orthogonality weight by a single barycentric factor. Although $\mathcal{K}$ preserves the total polynomial degree globally, it is *not* degree-preserving at the level of homogeneous subspaces.

More precisely, $\mathcal{K}$ satisfies the block filtering property

$$\mathcal{K} \mathcal{H}_j^D \subset \mathcal{H}_j^D \oplus \mathcal{H}_{j-1}^D, \quad j \geq 0,$$

with the convention $\mathcal{H}_{-1}^D = \{0\}$. Equivalently, the matrix of $\mathcal{K}$ in the graded basis

$$\Pi_n^D = \bigoplus_{j=0}^n \mathcal{H}_j^D$$

has a two-band block structure:

$$\mathcal{K} = \begin{pmatrix} A_0 & B_0 & 0 & 0 & \cdots \\ 0 & A_1 & B_1 & 0 & \cdots \\ 0 & 0 & A_2 & B_2 & \cdots \\ \vdots & & \ddots & \ddots & \ddots \end{pmatrix},$$

where

$$A_j : \mathcal{H}_j^D \rightarrow \mathcal{H}_j^D, \quad B_j : \mathcal{H}_{j+1}^D \rightarrow \mathcal{H}_j^D.$$

That is, writing u as a graded decomposition

$$u = \sum_{j=0}^n u_j, \quad u_j \in \mathcal{H}_j^{\kappa_{\text{src}}},$$

we define the operator

$$\mathcal{K} : \Pi_n^{\kappa_{\text{src}}} \rightarrow \Pi_n^{\kappa_{\text{tgt}}},$$

so that

$$\mathcal{K}u = \sum_{i=0}^n (\mathcal{K}u)_i, \quad (\mathcal{K}u)_i \in \mathcal{H}_i^{\text{tgt}}.$$

So, target block i receives contributions from source blocks i and $i+1$. Now, if u only has one homogeneous component, $u \equiv u_j \in \mathcal{H}_j^{\text{src}}$, then the only terms remaining in the sum over i from the formula

$$(\mathcal{K}u)_i = A_i u_i + B_i u_{i+1}$$

are $i = j$ via $A_j u_j$ and $i = j-1$ via $B_{j-1} u_j$. Then, we see that

$$\mathcal{K}u_j = (\mathcal{K}u)_j + (\mathcal{K}u)_{j-1} \Rightarrow \mathcal{K}\mathcal{H}_j \subset \mathcal{H}_j \oplus \mathcal{H}_{j-1}.$$

Sparsity assumption. There exist finite sets of multi-index offsets

$$\Delta_{\mathcal{K}}^{(0)}, \Delta_{\mathcal{K}}^{(-1)} \subset \mathbb{Z}^D,$$

independent of j , such that for every basis function $\phi_{\alpha}^{(\kappa_{\text{src}})}$ with $|\alpha| = j$,

$$\mathcal{K}\phi_{\alpha}^{(\kappa_{\text{src}})} \in \text{span}\left(\{\phi_{\alpha+\Delta}^{(\kappa_{\text{tgt}})} : \Delta \in \Delta_{\mathcal{K}}^{(0)}\} \cup \{\phi_{\alpha+\Delta}^{(\kappa_{\text{tgt}})} : \Delta \in \Delta_{\mathcal{K}}^{(-1)}, |\alpha+\Delta| = j-1\}\right).$$

The sets $\Delta_{\mathcal{K}}^{(0)}$ and $\Delta_{\mathcal{K}}^{(-1)}$ encode, respectively, the intra-degree couplings (A_j blocks) and the one-degree-down couplings (B_j blocks).

Stencil discovery. To determine these coupling patterns, we assemble a dense reference operator $\mathcal{K}^{(n_{\text{test}})}$ at a small test degree n_{test} and restrict it to the representative row block

$$\mathcal{K}_{\text{rep}}^{(n_{\text{test}})} = \mathcal{K}^{(n_{\text{test}})}|_{\mathcal{H}_{j_{\text{rep}}}^D \times (\mathcal{H}_{j_{\text{rep}}}^D \oplus \mathcal{H}_{j_{\text{rep}}+1}^D)}, \quad j_{\text{rep}} = n_{\text{test}} - 1.$$

For each nonzero entry in this restricted block, we record the multi-index difference

$$\Delta = \alpha_{\text{row}} - \alpha_{\text{col}},$$

partitioning the resulting offsets according to whether they correspond to $\mathcal{H}_j \leftarrow \mathcal{H}_j$ or $\mathcal{H}_j \leftarrow \mathcal{H}_{j+1}$ couplings. Once these offset sets stabilize as n_{test} increases, they are reused to assemble $\mathcal{K}$ at arbitrary polynomial degree n without further dense computation.

(iii) Three-term recurrence operators (JMat). Let

$$\mathcal{J} : \Pi_n^D \rightarrow \Pi_{n+1}^D$$

be a linear operator satisfying

$$\mathcal{J}\mathcal{H}_j^D \subset \mathcal{H}_{j-1}^D \oplus \mathcal{H}_j^D \oplus \mathcal{H}_{j+1}^D.$$

Then the matrix of $\mathcal{J}$ has block tridiagonal structure:

$$\mathcal{J} = \begin{pmatrix} B_0 & C_0 & 0 & \cdots & 0 \\ A_1 & B_1 & C_1 & \ddots & \vdots \\ 0 & A_2 & B_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & C_{n-1} \\ 0 & \cdots & 0 & A_n & B_n \end{pmatrix},$$

with

$$A_j : \mathcal{H}_j^D \rightarrow \mathcal{H}_{j-1}^D, \quad B_j : \mathcal{H}_j^D \rightarrow \mathcal{H}_j^D, \quad C_j : \mathcal{H}_j^D \rightarrow \mathcal{H}_{j+1}^D.$$

Sparsity assumption. There exists a finite set

$$\Delta_{\mathcal{J}} \subset \mathbb{Z}^D, \quad \sum_{d=1}^D \Delta_d \in \{-1, 0, +1\},$$

independent of j , such that

$$\mathcal{J}\phi_{\alpha}^{(\kappa)} \in \text{span}\{\phi_{\alpha+\Delta}^{(\kappa')} : \Delta \in \Delta_{\mathcal{J}}\}.$$

Delta-based representation of operator couplings

Let $\mathcal{O}$ be a linear operator acting on the graded polynomial space

$$\Pi_n^D = \bigoplus_{j=0}^n \mathcal{H}_j^D,$$

where $\mathcal{H}_j^D$ denotes the subspace of homogeneous polynomials of total degree j in D variables. Fix a Jacobi simplex basis $\{\phi_{\alpha}^{(\kappa)} : |\alpha| = j\}$ for each $\mathcal{H}_j^D$, indexed by multi-indices $\alpha \in \mathbb{N}^D$ with $|\alpha| = j$.

Coupling types. We assume that the action of $\mathcal{O}$ on a basis function $\phi_{\alpha}^{(\kappa)}$ can be written in the form

$$\mathcal{O}\phi_{\alpha}^{(\kappa)} = \sum_{\Delta \in \Delta_{\mathcal{O}}} c_{\Delta}(\alpha) \phi_{\alpha+\Delta}^{(\kappa')},$$

where:

- $\Delta_{\mathcal{O}} \subset \mathbb{Z}^D$ is a finite set of *multi-index shifts* (“deltas”),
- the coefficients $c_{\Delta}(\alpha)$ may depend on α and the Jacobi parameters, but
- the set $\Delta_{\mathcal{O}}$ itself is independent of the polynomial degree j .

Each $\Delta \in \Delta_{\mathcal{O}}$ represents a distinct *coupling type* between a source basis function indexed by α and a target basis function indexed by $\alpha + \Delta$.

Admissibility. For a fixed source multi-index α , a shift Δ is said to be *admissible* if the target multi-index

$$\beta = \alpha + \Delta$$

satisfies:

1. componentwise nonnegativity, $\beta_i \geq 0$ for all i , and
2. the appropriate degree constraint

$$|\beta| = |\alpha| + s,$$

where s is the degree shift associated with $\mathcal{O}$.

Only admissible shifts contribute to the action of the operator at a given α .

Stability of the delta set. For small degrees j , some coupling types may be absent due to boundary effects in the index set $\{\alpha : |\alpha| = j\}$. However, there exists a degree j_* such that for all $j \geq j_*$ the set of observed admissible shifts coincides with a fixed finite set $\Delta_{\mathcal{O}}$. We refer to this property as *stability* of the delta set.

Assembly at arbitrary degree. Once a stable delta set $\Delta_{\mathcal{O}}$ has been identified, the matrix representation of $\mathcal{O}$ on Π_n^D for arbitrary n may be constructed as follows:

1. loop over all basis functions (columns) indexed by α ,
2. for each α , loop over all $\Delta \in \Delta_{\mathcal{O}}$,
3. form the candidate target index $\beta = \alpha + \Delta$,
4. if β is admissible, insert the corresponding matrix entry coupling $\phi_{\alpha}^{(\kappa)}$ to $\phi_{\beta}^{(\kappa')}$.

The mapping between multi-indices and linear matrix indices is provided by the fixed ordering of the Jacobi basis within each homogeneous subspace and depends on the total degree, but the set of admissible shifts $\Delta_{\mathcal{O}}$ is independent of n .

Interpretation. In this formulation, the delta set $\Delta_{\mathcal{O}}$ encodes all possible local coupling types induced by the operator in multi-index space. The degree dependence of the matrix sparsity pattern arises solely from the admissibility conditions and the enumeration of basis functions, not from changes in the underlying operator structure. The apparent spreading of nonzeros with increasing degree is a consequence of the growing number of basis functions between two fixed multi-index shifts in the graded ordering; the underlying multi-index coupling encoded by each delta remains unchanged.

Data-driven discovery of the finite delta sets Δ_*

For each operator family $\mathcal{O}$ (e.g. $\mathcal{D}$, $\mathcal{K}$, $\mathcal{J}$) we assume the existence of a finite multi-index shift set

$$\Delta_{\mathcal{O}} \subset \mathbb{Z}^D$$

such that, for every homogeneous degree j and every basis index α with $|\alpha| = j$, the action of $\mathcal{O}$ on $\phi_{\alpha}^{(\kappa)}$ is supported only on shifts $\alpha + \Delta$ with $\Delta \in \Delta_{\mathcal{O}}$ (cf. the sparsity assumptions above). We describe a practical strategy to discover $\Delta_{\mathcal{O}}$.

Step 0: choose a representative homogeneous block. Fix a test degree n_{test} and select a representative block degree

$$j_{\text{rep}} = j_{\text{rep}}(n_{\text{test}})$$

for extracting the stencil. In our implementation we take

$$j_{\text{rep}} = n_{\text{test}} - 1,$$

i.e. the second highest-degree homogeneous block in $\Pi_{n_{\text{test}}}^D$, which is the largest block and empirically exhibits the full interior coupling.

Step 1: assemble a pruned dense test operator. Assemble a dense matrix representation

$$O^{(n_{\text{test}})} \in \mathbb{R}^{M \times M}, \quad M = \dim \Pi_{n_{\text{test}}}^D,$$

for the operator $\mathcal{O}$ (with the desired parameters κ , shifts, and axis), and apply numerical pruning to remove entries below a relative tolerance (e.g. row-relative threshold proportional to machine epsilon). This produces a matrix whose remaining nonzeros reflect the algebraic sparsity pattern.

Step 2: extract the set of observed deltas from the representative block. Restrict $O^{(n_{\text{test}})}$ to the representative homogeneous block

$$O_{\text{rep}}^{(n_{\text{test}})} = O^{(n_{\text{test}})}|_{\mathcal{H}_{j_{\text{rep}}}^D \times \mathcal{H}_{j_{\text{rep}}+s}^D},$$

where the degree shift s is determined by the operator:

$$s = \begin{cases} -1, & \mathcal{O} = \mathcal{D} \\ \in \{-1, 0\}, & \mathcal{O} = \mathcal{K} \\ \in \{-1, 0, +1\}, & \mathcal{O} = \mathcal{J} \text{ (handled per block band)} \end{cases}$$

and where $\mathcal{H}_{j_{\text{rep}}+s}^D$ denotes the appropriate source block. For each nonzero entry

$$(O_{\text{rep}}^{(n_{\text{test}})})_{i,j} \neq 0,$$

let α_i and α_j be the (unique) multi-indices associated with the row and column basis functions in the chosen ordering (existence assumed). Define the observed delta

$$\Delta = \alpha_i - \alpha_j \in \mathbb{Z}^D.$$

Collect all such deltas and retain the unique set:

$$\Delta_{\mathcal{O}}^{(n_{\text{test}})} = \left\{ \alpha_i - \alpha_j : (O_{\text{rep}}^{(n_{\text{test}})})_{i,j} \neq 0 \right\}.$$

Step 3: stabilization over increasing n_{test} . Starting from a minimal degree n_{min} (e.g. $n_{\text{min}} = 5$), repeat Steps 1–2 for $n_{\text{test}} = n_{\text{min}}, n_{\text{min}} + 1, \dots, n_{\text{max}}$, producing a sequence of candidate delta sets $\Delta_{\mathcal{O}}^{(n_{\text{test}})}$. We declare the stencil to have stabilized when

$$\Delta_{\mathcal{O}}^{(n_{\text{test}})} = \Delta_{\mathcal{O}}^{(n_{\text{test}}-1)}$$

(as sets, e.g. after sorting a canonical encoding), and set

$$\Delta_{\mathcal{O}} := \Delta_{\mathcal{O}}^{(n_{\text{test}})}.$$

Usage at arbitrary degree. Once $\Delta_{\mathcal{O}}$ is known, the operator at any target degree n can be assembled by looping over source basis functions (multi-indices α) and applying the admissible shifts:

$$\alpha \mapsto \alpha + \Delta, \quad \Delta \in \Delta_{\mathcal{O}},$$

discarding invalid targets (e.g. negative components or incorrect total degree), and placing the corresponding nonzero entry in the sparse matrix representation. The conversion between multi-indices and local indices within $\mathcal{H}_j^D$ is performed by fixed, degree-consistent maps induced by the chosen ordering.

13.1 Multiplication operators via Jacobi matrices and an operator-valued Clenshaw algorithm

Spaces and ordering. Fix $D \geq 1$, parameters $\kappa \in \mathbb{R}^{D+1}$ with $\kappa_r > -1/2$, and a degree cutoff $n \geq 0$. Let Π_n^D be the space of polynomials of total degree $\leq n$ on the simplex Δ^D with an *orthonormal* Jacobi simplex basis $\{\phi_\alpha^{(\kappa)}\}_{|\alpha| \leq n}$ ordered by total degree. Write

$$M_n := \dim \Pi_n^D, \quad \mathbb{R}^{M_n} \cong \bigoplus_{j=0}^n \mathbb{R}^{m_j}, \quad m_j := \dim \mathcal{H}_j^D,$$

and block any coefficient vector $c \in \mathbb{R}^{M_n}$ as $c = [c_0; \dots; c_n]$, $c_j \in \mathbb{R}^{m_j}$.

Degree embeddings/extractions. Let $E_j^{(n)} \in \mathbb{R}^{M_n \times m_j}$ embed a degree- j block into the global vector (zeros elsewhere), and $(E_j^{(n)})^T$ extract:

$$c_j = (E_j^{(n)})^T c, \quad c = \sum_{j=0}^n E_j^{(n)} c_j.$$

13.1.1 Coordinate Jacobi matrices: block structure and symmetry

For each coordinate x_i ($i = 1, \dots, D$) define the coordinate-multiplication (Jacobi) matrix

$$J_i^{(n)} \in \mathbb{R}^{M_n \times M_n}, \quad (J_i^{(n)})_{\ell k} = \langle \phi_\ell^{(\kappa)}, x_i \phi_k^{(\kappa)} \rangle_{w_\kappa}.$$

Because the basis is orthonormal and x_i is self-adjoint in $\langle \cdot, \cdot \rangle_{w_\kappa}$,

$$J_i^{(n)} = (J_i^{(n)})^T \quad (\text{real symmetric}).$$

Degree-tridiagonal structure. Multiplication by a coordinate changes total degree by at most 1, hence

$$(E_r^{(n)})^T J_i^{(n)} E_s^{(n)} = 0 \quad \text{unless} \quad r \in \{s-1, s, s+1\}.$$

Define the degree-coupling blocks:

$$\begin{aligned} U_{i,j}^{(n)} &:= (E_{j+1}^{(n)})^T J_i^{(n)} E_j^{(n)} \in \mathbb{R}^{m_{j+1} \times m_j}, \\ B_{i,j}^{(n)} &:= (E_j^{(n)})^T J_i^{(n)} E_j^{(n)} \in \mathbb{R}^{m_j \times m_j}, \\ L_{i,j}^{(n)} &:= (E_{j-1}^{(n)})^T J_i^{(n)} E_j^{(n)} \in \mathbb{R}^{m_{j-1} \times m_j}. \end{aligned}$$

By symmetry,

$$L_{i,j+1}^{(n)} = (U_{i,j}^{(n)})^T, \quad B_{i,j}^{(n)} = (B_{i,j}^{(n)})^T.$$

13.1.2 The multiplication operator $M_q^{(n)}$ as a matrix polynomial

Let $f \in \Pi_n^D$ have coefficient vector $c \in \mathbb{R}^{M_n}$:

$$f(x) = \sum_{|\alpha| \leq n} c_\alpha \phi_\alpha^{(\kappa)}(x).$$

Given a polynomial q (or a truncated Jacobi expansion) we define the *aliased multiplication operator*

$$\mathcal{M}_q^{(n)} := \Pi_n \circ (q \cdot) \circ \Pi_n : \Pi_n^D \rightarrow \Pi_n^D,$$

and its coefficient action is

$$y = M_q^{(n)} c, \quad y, c \in \mathbb{R}^{M_n},$$

where y are the coefficients of $\Pi_n(qf)$ in the same orthonormal basis.

Functional calculus definition. Since $J_i^{(n)}$ represent coordinate multiplication on Π_n^D (in coefficient space), we define

$$M_q^{(n)} := q(J_1^{(n)}, \dots, J_D^{(n)})$$

in the standard multivariate matrix-polynomial sense (well-defined because the $J_i^{(n)}$ commute in the infinite-dimensional setting, and in practice are consistent truncations of the same commuting operators; this is exactly the viewpoint adopted in the triangle case in [1]).

Key correction (dependence on c). Everything below constructs an *apply-only* algorithm for

$$y = M_q^{(n)} c,$$

i.e. the recurrence is driven by the input coefficient vector c (coefficients of f), not just by q .

13.1.3 Why a lifted space is sometimes necessary

If $\deg q = p$, then $\deg(qf) \leq n + p$ before projection back to Π_n^D . To avoid aliasing during intermediate steps, we introduce a *lift* degree $N \geq n + p$ and work on Π_N^D :

$$M_q^{(n)} c = (E^{(N \rightarrow n)})^T (M_q^{(N)} E^{(n \rightarrow N)} c),$$

where $E^{(n \rightarrow N)} : \mathbb{R}^{M_n} \rightarrow \mathbb{R}^{M_N}$ is the natural zero-padding embedding of coefficients and $(E^{(N \rightarrow n)})^T$ extracts degrees $\leq n$. In what follows we therefore describe the algorithm on Π_N^D (drop the superscript (N) for readability).

13.1.4 Stacked raising blocks and a left-inverse

For each degree j define the stacked *upward* block (compare the triangle B_n in [1])

$$\mathcal{B}_j := \begin{bmatrix} U_{1,j} \\ \vdots \\ U_{D,j} \end{bmatrix} \in \mathbb{R}^{(D m_{j+1}) \times m_j}.$$

Assume $\mathcal{B}_j$ has full column rank; define a left-inverse (minimum-norm)

$$\mathcal{B}_j^+ \in \mathbb{R}^{m_j \times (D m_{j+1})}, \quad \mathcal{B}_j^+ \mathcal{B}_j = I_{m_j}.$$

In code, $\mathcal{B}_j^+$ is applied via cached sparse QR factors of $\mathcal{B}_j$.

13.1.5 OTV-style operator-valued Clenshaw: the Kronecker formulation

Coefficient blocks of q . Represent q in the *same orthonormal simplex basis* on Π_p^D :

$$q(x) = \sum_{j=0}^p \Phi_j(x)^T q_j, \quad q_j \in \mathbb{R}^{m_j},$$

where $\Phi_j(x) \in \mathbb{R}^{m_j}$ stacks the degree- j basis functions.

The central identity (triangle analogue (2.12)). The operator-valued Clenshaw algorithm is most cleanly stated as a *triangular block solve* in a lifted product space, exactly as in [1, §2.4]. Define unknowns

$$v_j \in \mathbb{R}^{M_N}, \quad j = 0, 1, \dots, p,$$

and stack $v := [v_0; v_1; \dots; v_p] \in \mathbb{R}^{(p+1)M_N}$.

Define a lower block-triangular operator $\tilde{L}_p(J)$ such that the formal “evaluation” vector $(\Phi_0(J), \dots, \Phi_p(J))$ satisfies

$$\tilde{L}_p(J) \begin{bmatrix} \Phi_0(J) \\ \Phi_1(J) \\ \vdots \\ \Phi_p(J) \end{bmatrix} = e_0 \otimes I,$$

where e_0 is the first canonical basis vector in $\mathbb{R}^{p+1}$ and I is the identity on $\mathbb{R}^{M_N}$. Then the multiplication operator action applied to c is

$$y = M_q^{(N)} c = (e_0^T \otimes I) \left(\tilde{L}_p(J)^{-T} (q \otimes c) \right). \quad (22)$$

This is the exact analogue of the triangle formula $M_q = (e_0 \otimes I) L^{-T} q$ in [1, (??)], except that we explicitly apply it to c by replacing q with $q \otimes c$.

Interpretation. Equation (22) says:

- build a block upper-triangular system $\tilde{L}_p(J)^T v = q \otimes c$;
- solve it by back substitution (this *is* operator-valued Clenshaw);
- return $y = v_0$.

Thus the dependence on c is explicit and unavoidable: the right-hand side is $q \otimes c$.

13.1.6 Making $\tilde{L}_p(J)$ explicit using only SpMV with J_i and $\mathcal{B}_j^+$

We now express the back-substitution update using only: (i) sparse SpMV with J_i on $\mathbb{R}^{M_N}$, and (ii) applications of $\mathcal{B}_j^+$ on stacked degree blocks.

Degree extract/stack operators. For any $w \in \mathbb{R}^{M_N}$ define $w^{(j)} := (E_j^{(N)})^T w \in \mathbb{R}^{m_j}$. Define the stacked degree extraction

$$\mathcal{E}_{j+1}(w) := \begin{bmatrix} (E_{j+1}^{(N)})^T (J_1 w) \\ \vdots \\ (E_{j+1}^{(N)})^T (J_D w) \end{bmatrix} \in \mathbb{R}^{Dm_{j+1}}.$$

The two-step Clenshaw states. Define $b_{p+1} = 0$, $b_{p+2} = 0$ in $\mathbb{R}^{M_N}$. For $j = p, p-1, \dots, 0$, compute

$$b_j = \underbrace{\left(E_j^{(N)} q_j\right) \ll \cdot \gg c}_{\text{degree-}j \text{ forcing from } q \otimes c} + \underbrace{E_j^{(N)} \left(\mathcal{B}_j^+ \mathcal{E}_{j+1}(b_{j+1})\right)}_{\text{folded action of } J_i \text{ on } b_{j+1}} - \underbrace{E_j^{(N)} \left(\mathcal{C}_{j+1} (E_{j+1}^{(N)})^T b_{j+2}\right)}_{\text{two-step correction}}, \quad (23)$$

and return $y = b_0$ (and finally restrict degrees $\leq n$ if you lifted).

What is the contraction $\ll \cdot \gg$? (This fixes the earlier mistake.) In (22), the right-hand side is $q \otimes c$, not an entrywise product. Accordingly, the correct “forcing” in (23) is:

$$\left(E_j^{(N)} q_j\right) \ll \cdot \gg c \equiv (E_j^{(N)} \otimes I) (q_j \otimes c) = E_j^{(N)} q_j \otimes c,$$

i.e. it is *Kronecker injection*, not Hadamard multiplication. In implementation you never explicitly form $q_j \otimes c$ as a dense tensor; rather, you run the back substitution on block-vectors in $\mathbb{R}^{M_N}$ with the correct RHS assembly: each degree- j block RHS is q_j *times* the current c in the Kronecker sense. (For testing, you *can* form $q \otimes c$ explicitly for small sizes to validate the recurrence.)

The correction operator $\mathcal{C}_{j+1}$. The last term in (23) is the analogue of the $-\beta_{j+1} b_{j+2}$ term in 1D. In OTV’s triangle setting it is induced by the next-level coupling after applying the left-inverse (see the discussion surrounding the left-inverse B_n^+). For the D -simplex, the clean way to define it is:

$$\mathcal{C}_{j+1} := \mathcal{B}_j^+ \begin{bmatrix} B_{1,j+1} \\ \vdots \\ B_{D,j+1} \end{bmatrix} \in \mathbb{R}^{m_j \times m_{j+1}} \quad (\text{same-degree leakage folded down}).$$

Then the two-step correction uses b_{j+2} through its degree- $(j+1)$ block:

$$E_j^{(N)} \left(\mathcal{C}_{j+1} (E_{j+1}^{(N)})^T b_{j+2}\right).$$

This is precisely the piece that an “ad hoc down-block subtraction” gets wrong in $D > 1$ if you do not track the *same-degree* leakage through $B_{i,j+1}$ before applying $\mathcal{B}_j^+$.

Returned quantity. After the sweep, set

$$y^{(N)} := b_0 \in \mathbb{R}^{M_N}, \quad y^{(n)} := (E^{(N \rightarrow n)})^T y^{(N)} \in \mathbb{R}^{M_n}.$$

Then $y^{(n)}$ are exactly the coefficients of $\Pi_n(qf)$.

13.1.7 Worked schematic for $p = 4$ (showing the dependence on c)

Let $p = 4$, initialize $b_6 = b_5 = 0$ and compute $b_4, \dots, b_0$ using (23):

$$\begin{aligned} b_4 &= \left(E_4 q_4\right) \ll \cdot \gg c, \\ b_3 &= \left(E_3 q_3\right) \ll \cdot \gg c + E_3 \left(\mathcal{B}_3^+ \mathcal{E}_4(b_4)\right), \\ b_2 &= \left(E_2 q_2\right) \ll \cdot \gg c + E_2 \left(\mathcal{B}_2^+ \mathcal{E}_3(b_3)\right) - E_2 \left(\mathcal{C}_3 (E_3)^T b_4\right), \\ b_1 &= \left(E_1 q_1\right) \ll \cdot \gg c + E_1 \left(\mathcal{B}_1^+ \mathcal{E}_2(b_2)\right) - E_1 \left(\mathcal{C}_2 (E_2)^T b_3\right), \\ b_0 &= \left(E_0 q_0\right) \ll \cdot \gg c + E_0 \left(\mathcal{B}_0^+ \mathcal{E}_1(b_1)\right) - E_0 \left(\mathcal{C}_1 (E_1)^T b_2\right), \quad y = b_0. \end{aligned}$$

Every line depends on c via the RHS forcing and through the propagated b_j .

References

- [1] S. Olver, A. Townsend, and G. Vasil, *A sparse spectral method on triangles*, arXiv:1902.04863, 2019.